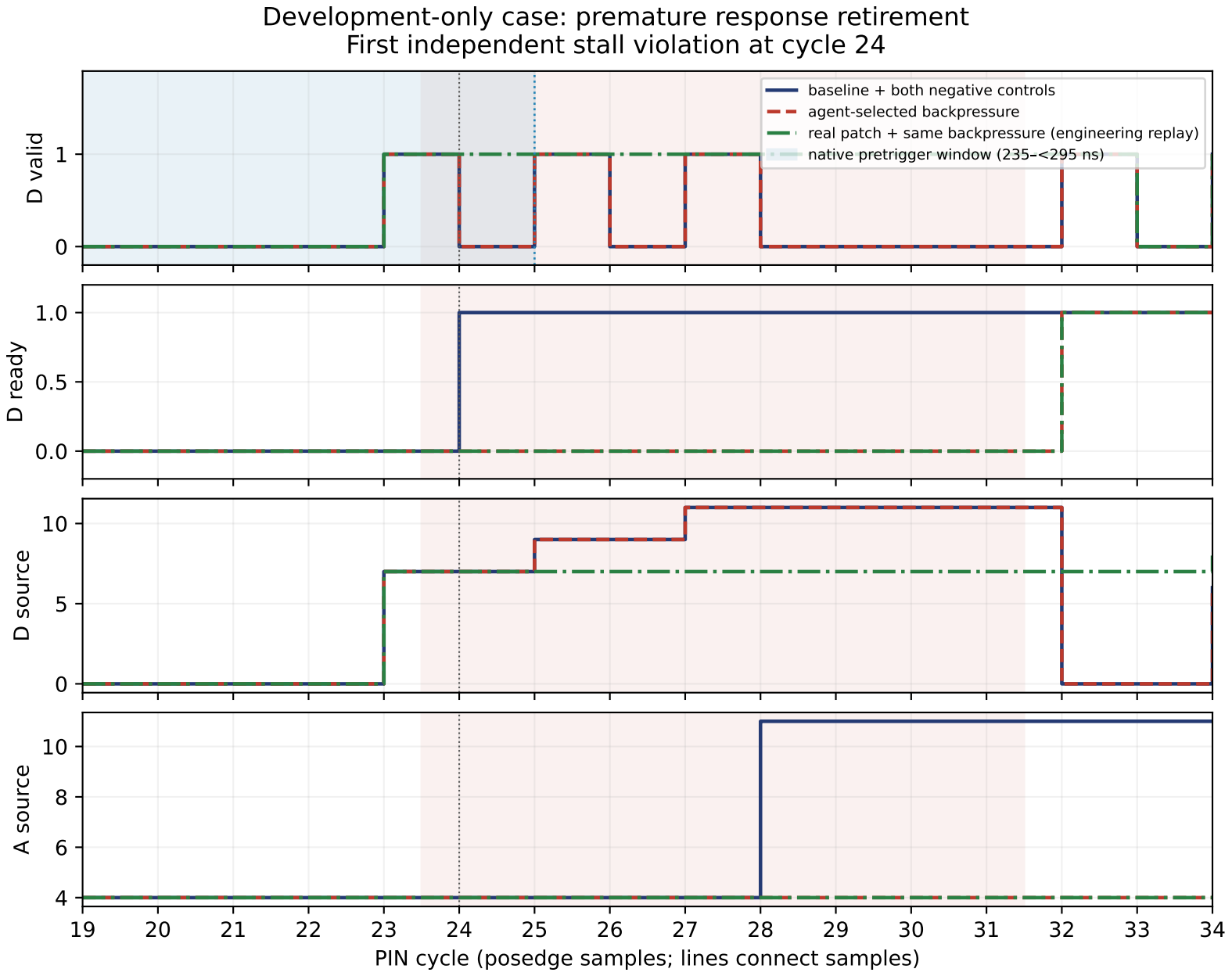
Development-only case: premature response retirement
First independent stall violation at cycle 24
baseline + both negative controls
agent-selected backpressure
real patch + same backpressure (engineering replay)
native pretrigger window (235–<295 ns)
D valid
D ready
D source
A source
PIN cycle (posedge samples; lines connect samples)